Healthcare_Expenditure_in_aging_era

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Import libary

```r
library(readxl)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(stringr)
library(plm)
```

```
##
## Attaching package: 'plm'

## The following objects are masked from 'package:dplyr':
##
##     between, lag, lead
```

```r
library(ggplot2)
```

Read Data

```r
ghed_raw <- read_excel("/Users/ez_us/Documents/Aging_HCE/datasets/GHED_data.XLSX", sheet= "Data")
```

Check the variable names

```r
str(ghed_raw)
```

```
## tibble [4,612 x 4,120] (S3: tbl_df/tbl/data.frame)
##  $ location                  : chr [1:4612] "Algeria" "Algeria" "Algeria" "Algeria" ...
##  $ code                      : chr [1:4612] "DZA" "DZA" "DZA" "DZA" ...
##  $ region                    : chr [1:4612] "AFR" "AFR" "AFR" "AFR" ...
##  $ income                    : chr [1:4612] "Upper-middle" "Upper-middle" "Upper-middle" "Upper-middle
##  $ year                      : num [1:4612] 2000 2001 2002 2003 2004 ...
##  $ che_gdp                   : num [1:4612] 3.21 3.54 3.44 3.33 3.29 ...
```

```
##  $ che_pc_usd             : num [1:4612] 61.9 67.1 66.7 76 92.7 ...
##  $ che                    : num [1:4612] 143870 162231 168702 189137 217929 ...
##  $ gghed                  : num [1:4612] 103534 123664 126997 145057 155500 ...
##  $ pvtd                   : num [1:4612] 40261 38492 41630 43985 62327 ...
##  $ ext                    : num [1:4612] 75.1 75.1 75.1 95 102 ...
##  $ dom_che                : num [1:4612] 99.9 100 100 99.9 100 ...
##  $ gghed_che              : num [1:4612] 72 76.2 75.3 76.7 71.4 ...
##  $ pvtd_che               : num [1:4612] 28 23.7 24.7 23.3 28.6 ...
##  $ oops_che               : num [1:4612] 25.8 21.7 22.5 21.1 26.1 ...
##  $ vpp_che                : num [1:4612] 0.818 0.848 0.954 0.981 1.445 ...
##  $ ext_che                : num [1:4612] 0.0522 0.0463 0.0445 0.0502 0.0468 ...
##  $ gghed_gdp              : num [1:4612] 2.31 2.7 2.59 2.55 2.35 ...
##  $ gghed_gge              : num [1:4612] 8.78 9.27 7.97 9.44 8.67 ...
##  $ gghed_pc_usd           : num [1:4612] 44.5 51.1 50.2 58.3 66.1 ...
##  $ pvtd_pc_usd            : num [1:4612] 17.3 15.9 16.5 17.7 26.5 ...
##  $ oop_pc_usd             : num [1:4612] 16 14.5 15 16.1 24.2 ...
##  $ ext_pc_usd             : num [1:4612] 0.0323 0.031 0.0297 0.0381 0.0434 ...
##  $ tran_shi               : num [1:4612] 0 0 0 0 0 0 0 0 0 0 ...
##  $ shise_shi              : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ cfa_che                : num [1:4612] 72 76.2 75.3 76.7 71.4 ...
##  $ gfa_che                : num [1:4612] 45.9 50.5 49.3 50.7 47.1 ...
##  $ chi_che                : num [1:4612] 26.1 25.8 25.9 26 24.2 ...
##  $ shi_che                : num [1:4612] 26.1 25.8 25.9 26 24.2 ...
##  $ chi_pvt_che            : num [1:4612] 0 0 0 0 0 0 0 0 0 0 ...
##  $ vfa_che                : num [1:4612] 28 23.8 24.7 23.3 28.6 ...
##  $ vhi_che                : num [1:4612] 0.818 0.848 0.954 0.981 1.445 ...
##  $ row_che                : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ phc_usd_pc             : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ phc_che                : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ phc_gghed_usd_pc       : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ gghed_phc_gghed        : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ gghed_phc_phc          : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ phc_ext_usd_pc         : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ ext_phc_ext            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ ext_phc_phc            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ phc_public_gdp         : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ phc_pvtd_usd_pc        : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ pvtd_phc_pvtd          : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ pvtd_phc_phc           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ hc62_che               : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ hc62_g_gghed           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ hc62_ext_ext           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ fp3214_che             : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ fp3214_gghed_fp3214    : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ fp3214_ext_fp3214      : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis1_che               : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis11_che              : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis12_che              : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis13_che              : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis16_che              : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis2_che               : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis21_che              : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis22_che              : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis23_che              : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
```

```
##  $ dis3_che                : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis4_che                : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis5_che                : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ disnec_che              : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis1_g_gghed            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis11_g_gghed           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis12_g_gghed           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis13_g_gghed           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis16_g_gghed           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis2_g_gghed            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis21_g_gghed           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis22_g_gghed           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis23_g_gghed           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis3_g_gghed            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis4_g_gghed            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis5_g_gghed            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ disnec_g_gghed          : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis1_ext_ext            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis11_ext_ext           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis12_ext_ext           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis13_ext_ext           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis16_ext_ext           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis2_ext_ext            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis21_ext_ext           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis22_ext_ext           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis23_ext_ext           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis3_ext_ext            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis4_ext_ext            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ dis5_ext_ext            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ disnec_ext_ext          : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ hccov_che               : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ hccov_usd_pc            : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ hccov_gghed_gghed       : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ hccov_ext_ext           : num [1:4612] NA NA NA NA NA NA NA NA NA NA ...
##  $ gge_gdp                 : num [1:4612] 26.4 29.1 32.5 27 27.1 ...
##  $ gdp_pc_usd              : num [1:4612] 1924 1896 1937 2284 2817 ...
##  $ fs                      : num [1:4612] 143870 162231 168702 189137 217929 ...
##  $ fs1                     : num [1:4612] 66051 81873 83241 95906 102685 ...
##  $ fs11                    : num [1:4612] 66051 81873 83241 95906 102685 ...
##   [list output truncated]
```

Select relevant GHED variables

```
ghed_clean <- ghed_raw %>%
  select(
    location,
    code,
    year,
    gghed_che,
    shi_che,
    oops_che,
    vhi_che
  ) %>%
```

```r
  filter(year >= 2000 & year <= 2023) %>%
  mutate(
    year = as.integer(year)
  )
```

Sanity check

```r
summary(ghed_clean)
```

```
##    location             code                year         gghed_che
##  Length:4590        Length:4590        Min.   :2000   Min.   : 0.786
##  Class :character   Class :character   1st Qu.:2006   1st Qu.:33.635
##  Mode  :character   Mode  :character   Median :2012   Median :52.039
##                                        Mean   :2012   Mean   :50.910
##                                        3rd Qu.:2018   3rd Qu.:69.868
##                                        Max.   :2023   Max.   :99.877
##                                                       NA's   :2
##     shi_che          oops_che            vhi_che
##  Min.   : 0.000   Min.   : 0.04459   Min.   : 0.0000
##  1st Qu.: 0.000   1st Qu.:16.44933   1st Qu.: 0.7441
##  Median : 1.616   Median :30.89642   Median : 2.6066
##  Mean   :14.896   Mean   :33.17211   Mean   : 4.9004
##  3rd Qu.:23.591   3rd Qu.:46.93554   3rd Qu.: 6.4341
##  Max.   :84.038   Max.   :87.90394   Max.   :50.6225
##  NA's   :174      NA's   :2          NA's   :422
```

```r
colSums(is.na(ghed_clean))
```

```
##  location      code      year gghed_che   shi_che  oops_che   vhi_che
##         0         0         0         2       174         2       422
```

Read World Bank aging data

```r
aging_raw <- read_excel(
  "/Users/ez_us/Documents/Aging_HCE/datasets/API_SP.POP.65UP.TO.ZS_DS2_en_excel_v2_1001.xls",
  sheet = "Data",
  skip = 3
)
```

Reshape Wolrd Bank data (wide to long)

```r
aging_long <- aging_raw %>%
  pivot_longer(
    cols = matches("^\\d{4}$"),
    names_to = "year",
    values_to = "aging_65"
  ) %>%
  rename(
    location = `Country Name`,
    code = `Country Code`
  ) %>%
  mutate(
    year = as.integer(year)
```

```r
  ) %>%
  filter(year >= 2000 & year <= 2023)
```

Check

```r
str(aging_long)
```

```
## tibble [6,384 x 6] (S3: tbl_df/tbl/data.frame)
##  $ location      : chr [1:6384] "Aruba" "Aruba" "Aruba" "Aruba" ...
##  $ code          : chr [1:6384] "ABW" "ABW" "ABW" "ABW" ...
##  $ Indicator Name: chr [1:6384] "Population ages 65 and above (% of total population)" "Population ag
##  $ Indicator Code: chr [1:6384] "SP.POP.65UP.TO.ZS" "SP.POP.65UP.TO.ZS" "SP.POP.65UP.TO.ZS" "SP.POP.6
##  $ year          : int [1:6384] 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 ...
##  $ aging_65      : num [1:6384] 6.78 7.02 7.28 7.55 7.84 ...
```

```r
summary(aging_long$aging_65)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##  0.8625  3.4902  5.9459  8.1166 12.1116 37.3193      24
```

Check for coverage gaps,which helps later when deciding which countries to keep.

```r
aging_long %>%
  group_by(location) %>%
  summarise(n_years = n()) %>%
  arrange(n_years)
```

```
## # A tibble: 266 x 2
##    location                    n_years
##    <chr>                         <int>
##  1 Afghanistan                      24
##  2 Africa Eastern and Southern      24
##  3 Africa Western and Central       24
##  4 Albania                          24
##  5 Algeria                          24
##  6 American Samoa                   24
##  7 Andorra                          24
##  8 Angola                           24
##  9 Antigua and Barbuda              24
## 10 Arab World                       24
## # i 256 more rows
```

Merge GHED + World Bank

```r
panel_data <- ghed_clean %>%
  left_join(aging_long, by = c("code", "year"))
```

Check result

```r
dim(panel_data)
```

```
## [1] 4590   11
```

```r
summary(panel_data)
```

```
##   location.x            code                year         gghed_che     
##  Length:4590        Length:4590        Min.   :2000   Min.   : 0.786  
##  Class :character   Class :character   1st Qu.:2006   1st Qu.:33.635  
##  Mode  :character   Mode  :character   Median :2012   Median :52.039  
##                                        Mean   :2012   Mean   :50.910  
##                                        3rd Qu.:2018   3rd Qu.:69.868  
##                                        Max.   :2023   Max.   :99.877  
##                                                       NA's   :2       
##     shi_che          oops_che           vhi_che          location.y       
##  Min.   : 0.000   Min.   : 0.04459   Min.   : 0.0000   Length:4590       
##  1st Qu.: 0.000   1st Qu.:16.44933   1st Qu.: 0.7441   Class :character  
##  Median : 1.616   Median :30.89642   Median : 2.6066   Mode  :character  
##  Mean   :14.896   Mean   :33.17211   Mean   : 4.9004                     
##  3rd Qu.:23.591   3rd Qu.:46.93554   3rd Qu.: 6.4341                     
##  Max.   :84.038   Max.   :87.90394   Max.   :50.6225                     
##  NA's   :174      NA's   :2          NA's   :422                         
##  Indicator Name     Indicator Code        aging_65      
##  Length:4590        Length:4590        Min.   : 0.8625  
##  Class :character   Class :character   1st Qu.: 3.3762  
##  Mode  :character   Mode  :character   Median : 5.4853  
##                                        Mean   : 8.0289  
##                                        3rd Qu.:12.3636  
##                                        Max.   :37.3193  
##                                        NA's   :48       
```

Convert to panel-data format (for models)

```r
panel_data <- pdata.frame(
  panel_data,
  index = c("code", "year")
)
```

First baseline models (run separately)

Example 1: Out-of-pocket share

```r
model_oops <- plm(
  oops_che ~ aging_65,
  data = panel_data,
  model = "within"
)

summary(model_oops)
```

```
## Oneway (individual) effect Within Model
## 
## Call:
## plm(formula = oops_che ~ aging_65, data = panel_data, model = "within")
## 
## Unbalanced Panel: n = 193, T = 6-24, N = 4540
## 
## Residuals:
```

```
##     Min.  1st Qu.   Median  3rd Qu.     Max.
## -26.7611  -2.9114  -0.1233   2.9257  31.2729
##
## Coefficients:
##           Estimate Std. Error t-value  Pr(>|t|)
## aging_65 -0.930569   0.071558 -13.005 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    174470
## Residual Sum of Squares: 167940
## R-Squared:      0.037456
## Adj. R-Squared: -0.0052897
## F-statistic: 169.116 on 1 and 4346 DF, p-value: < 2.22e-16
```

Goverment financing share

```r
model_gov <- plm(
  gghed_che ~ aging_65,
  data = panel_data,
  model = "within"
)

summary(model_gov)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = gghed_che ~ aging_65, data = panel_data, model = "within")
##
## Unbalanced Panel: n = 193, T = 6-24, N = 4540
##
## Residuals:
##      Min.   1st Qu.    Median   3rd Qu.      Max.
## -41.54853  -3.58454  -0.18273   3.51725  28.67543
##
## Coefficients:
##          Estimate Std. Error t-value  Pr(>|t|)
## aging_65 0.925974   0.076719   12.07 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    199510
## Residual Sum of Squares: 193040
## R-Squared:      0.032433
## Adj. R-Squared: -0.010536
## F-statistic: 145.677 on 1 and 4346 DF, p-value: < 2.22e-16
```

Social health insuramce

```r
model_shi <- plm(
  shi_che ~ aging_65,
  data = panel_data,
  model = "within"
```

```r
)

summary(model_shi)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = shi_che ~ aging_65, data = panel_data, model = "within")
##
## Unbalanced Panel: n = 192, T = 1-24, N = 4368
##
## Residuals:
##        Min.     1st Qu.      Median     3rd Qu.        Max.
## -51.4329636  -0.8289345  -0.0020622   0.6710956  45.9629845
##
## Coefficients:
##          Estimate Std. Error t-value  Pr(>|t|)
## aging_65 0.503214   0.051801  9.7143 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    85512
## Residual Sum of Squares: 83622
## R-Squared:      0.022103
## Adj. R-Squared: -0.022868
## F-statistic: 94.3674 on 1 and 4175 DF, p-value: < 2.22e-16
```

Voluntary health insurance

```r
model_vhi <- plm(
  vhi_che ~ aging_65,
  data = panel_data,
  model = "within"
)

summary(model_vhi)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = vhi_che ~ aging_65, data = panel_data, model = "within")
##
## Unbalanced Panel: n = 183, T = 2-24, N = 4123
##
## Residuals:
##       Min.    1st Qu.     Median    3rd Qu.       Max.
## -20.504737  -0.590593  -0.023784   0.458346  15.794670
##
## Coefficients:
##          Estimate Std. Error t-value Pr(>|t|)
## aging_65 0.020686   0.028774  0.7189   0.4722
##
## Total Sum of Squares:    22045
## Residual Sum of Squares: 22042
```

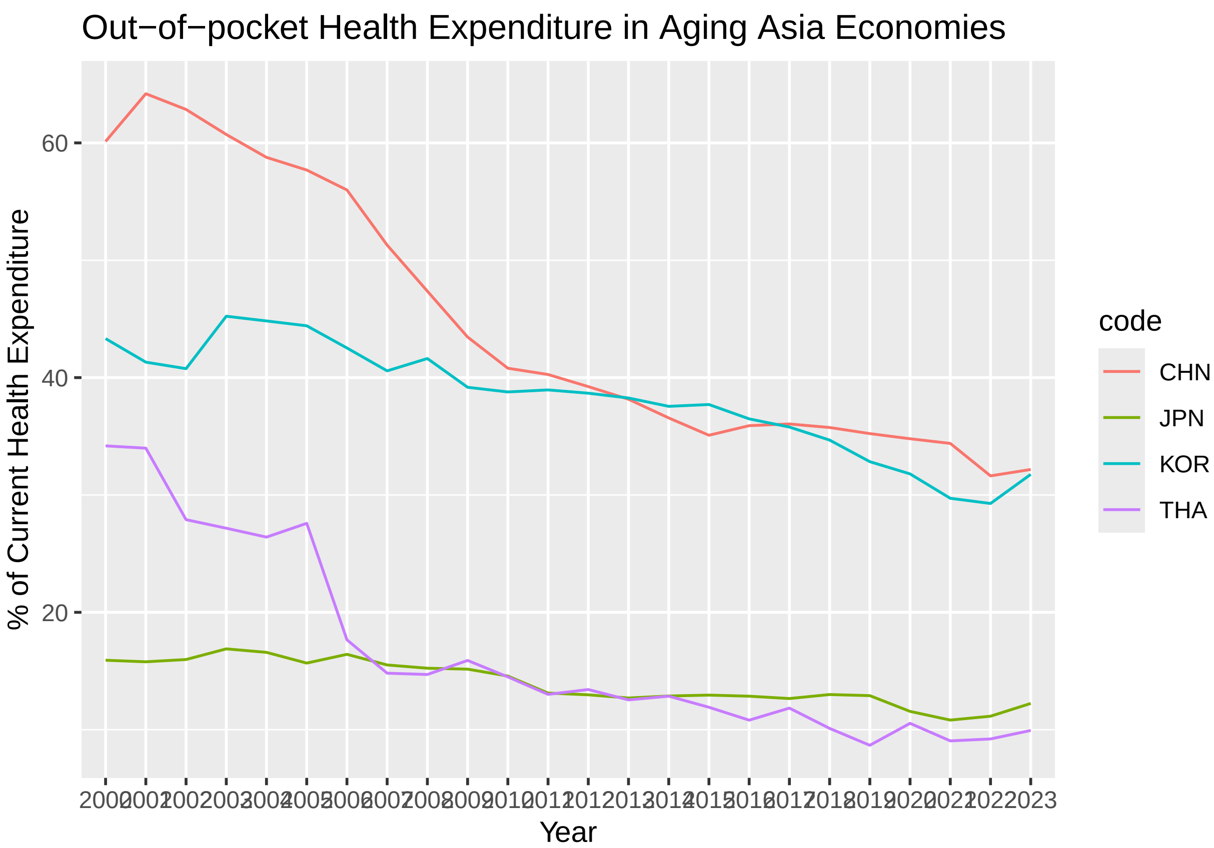
```
## R-Squared:      0.00013119
## Adj. R-Squared: -0.046321
## F-statistic: 0.516815 on 1 and 3939 DF, p-value: 0.47225
```

Asia-focused descriptive check(bonus)

```r
asia_codes <- c("CHN", "JPN", "KOR", "THA")

asia_data <- panel_data %>%
  filter(code %in% asia_codes) %>%
  as.data.frame()

asia_data <- pdata.frame(
  asia_data,
  index = c("code", "year")
)
```

test - generate oop plot

```r
ggplot(asia_data, aes(x = year, y = oops_che, color = code, group = code)) +
  geom_line() +
  labs(
  title = "Out-of-pocket Health Expenditure in Aging Asia Economies",
  x = "Year",
  y = "% of Current Health Expenditure"
)
```



Run financing models again

Example 1: Out-of-pocket share

```r
model_asia_oops <- plm(
  oops_che ~ aging_65,
  data = asia_data,
  model = "within"
)

summary(model_asia_oops)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = oops_che ~ aging_65, data = asia_data, model = "within")
##
## Balanced Panel: n = 4, T = 24, N = 96
##
## Residuals:
##     Min.  1st Qu.   Median  3rd Qu.     Max.
## -8.88300 -4.04747 -0.82188  2.30521 15.89507
##
## Coefficients:
##          Estimate Std. Error t-value  Pr(>|t|)
## aging_65 -1.52261    0.19004 -8.0119 3.619e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    4955.6
## Residual Sum of Squares: 2905.8
## R-Squared:      0.41362
## Adj. R-Squared: 0.38785
## F-statistic: 64.1904 on 1 and 91 DF, p-value: 3.6188e-12
```

Interpretation

> Holding country-specific characteristics constant, a one-percentage-point increase in the population aged 65+ is associated with a 1.5 percentage-point decline in the out-of-pocket share of health expenditure in Asian economies.

Example 2: Government financing share

```r
model_asia_gov <- plm(
  gghed_che ~ aging_65,
  data = asia_data,
  model = "within"
)

summary(model_asia_gov)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = gghed_che ~ aging_65, data = asia_data, model = "within")
```

```
##
## Balanced Panel: n = 4, T = 24, N = 96
##
## Residuals:
##      Min.   1st Qu.    Median   3rd Qu.      Max.
## -21.39120  -2.83413   0.17091   4.45948  13.18839
##
## Coefficients:
##          Estimate Std. Error t-value  Pr(>|t|)
## aging_65  1.19440    0.22167  5.3883 5.545e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    5214.6
## Residual Sum of Squares: 3953.3
## R-Squared:      0.24188
## Adj. R-Squared: 0.20856
## F-statistic: 29.0338 on 1 and 91 DF, p-value: 5.5451e-07
```

Interpretation

> Aging is associated with a statistically significant increase in the government share of health expenditure.
>
> This indicates: - Fiscal absorption of aging-related costs - Active public policy response - Alignment with UHC expansion trajectories

This result explains where the OOP reduction goes. ### Example 3: Social health insurance

```
model_asia_shi <- plm(
  shi_che ~ aging_65,
  data = asia_data,
  model = "within"
)

summary(model_asia_shi)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = shi_che ~ aging_65, data = asia_data, model = "within")
##
## Balanced Panel: n = 4, T = 24, N = 96
##
## Residuals:
##      Min.   1st Qu.    Median   3rd Qu.      Max.
## -17.21057  -0.90758   0.27254   1.65227  11.26353
##
## Coefficients:
##          Estimate Std. Error t-value  Pr(>|t|)
## aging_65  0.74175    0.17240  4.3026 4.241e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    2877.7
## Residual Sum of Squares: 2391.2
```

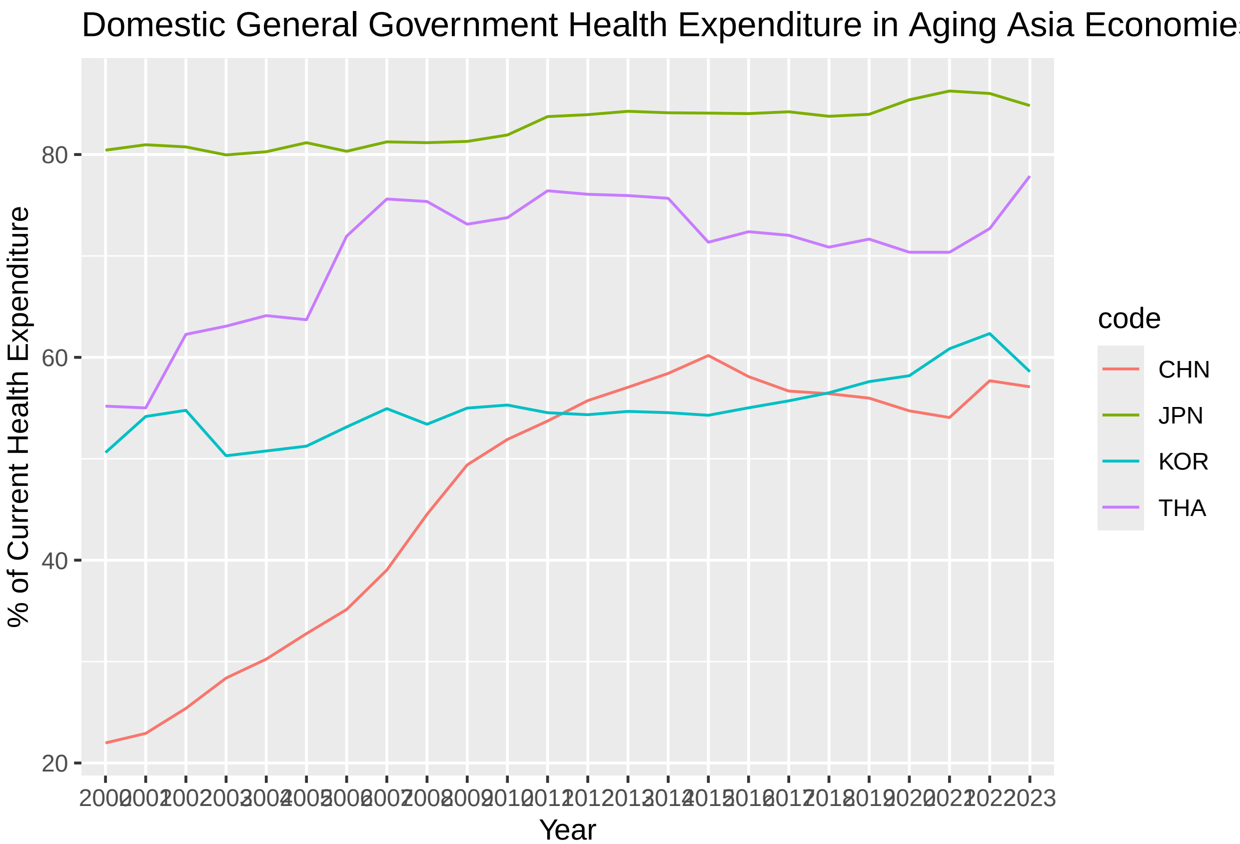
```
## R-Squared:      0.16904
## Adj. R-Squared: 0.13252
## F-statistic: 18.5125 on 1 and 91 DF, p-value: 4.2409e-05
```

Example 4: Voluntary health insurance

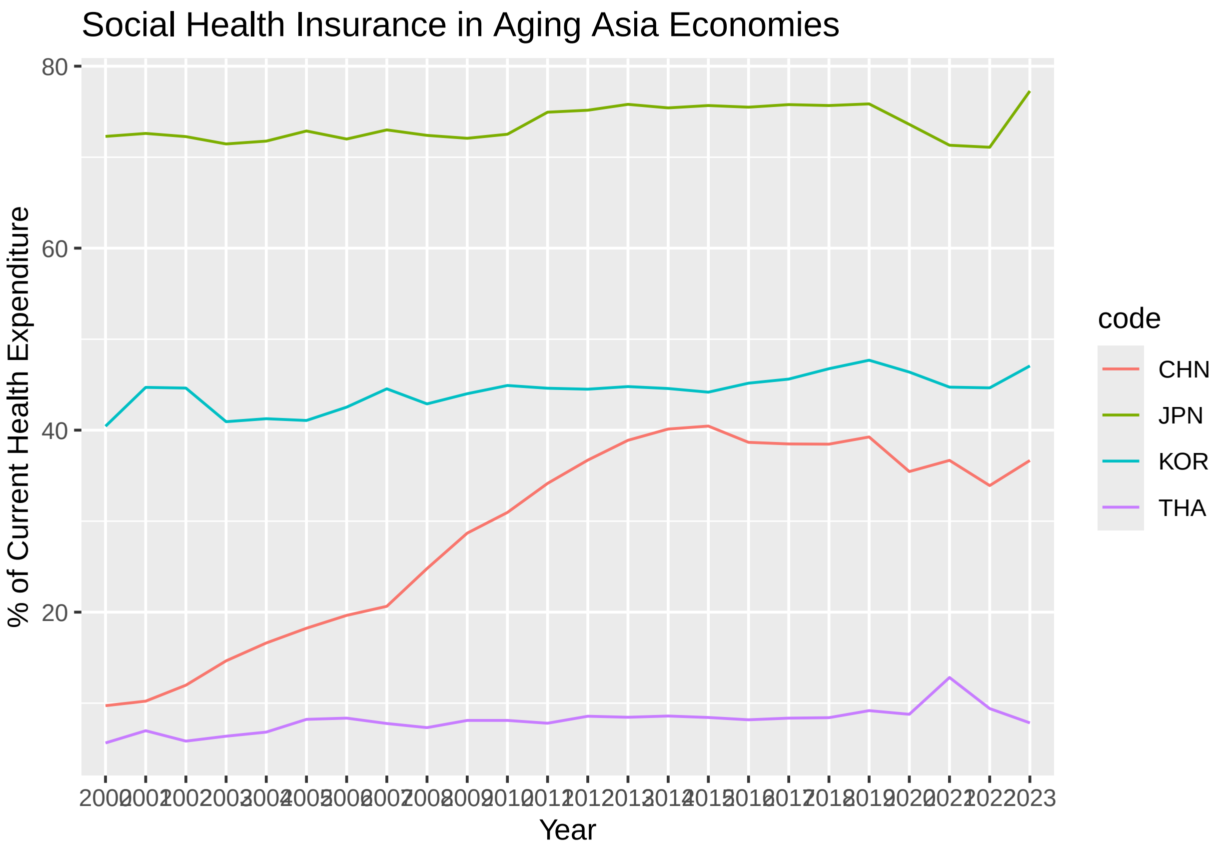
```r
model_asia_vhi <- plm(
  vhi_che ~ aging_65,
  data = asia_data,
  model = "within"
)

summary(model_asia_vhi)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = vhi_che ~ aging_65, data = asia_data, model = "within")
##
## Balanced Panel: n = 4, T = 24, N = 96
##
## Residuals:
##     Min.  1st Qu.   Median  3rd Qu.     Max.
## -3.36397 -1.38153 -0.42125  1.19861  6.78087
##
## Coefficients:
##          Estimate Std. Error t-value  Pr(>|t|)
## aging_65 0.579874   0.073139  7.9284 5.387e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    727.69
## Residual Sum of Squares: 430.39
## R-Squared:      0.40855
## Adj. R-Squared: 0.38255
## F-statistic: 62.8591 on 1 and 91 DF, p-value: 5.3871e-12
```

```r
ggplot(asia_data, aes(x = year, y = gghed_che, color = code, group = code)) +
  geom_line() +
  labs(
  title = "Domestic General Government Health Expenditure in Aging Asia Economies",
  x = "Year",
  y = "% of Current Health Expenditure"
)
```




```r
ggplot(asia_data, aes(x = year, y = shi_che, color = code, group = code)) +
  geom_line() +
  labs(
  title = "Social Health Insurance in Aging Asia Economies",
  x = "Year",
  y = "% of Current Health Expenditure"
)
```



Core question: "Is the effect stronger in countries that are actually aging rapidly?"

This tests whether your results are mechanically driven by aging, not by global trends.

Calculate country-level aging change

```
aging_intensity <- panel_data %>%
  filter(year %in% c(2000, 2023)) %>%
  select(code, year, aging_65) %>%
  group_by(code) %>%
  filter(n() == 2) %>%              # KEEP only countries with both years
  ungroup() %>%
  pivot_wider(
    names_from = year,
    values_from = aging_65
  ) %>%
  mutate(
    aging_change = `2023` - `2000`
  )
```

High-aging vs low-aging (median split)

```
median_change <- median(aging_intensity$aging_change, na.rm = TRUE)
```

```r
aging_intensity <- aging_intensity %>%
  mutate(
    high_aging = ifelse(aging_change >= median_change, 1, 0)
  )

panel_data <- as.data.frame(panel_data)

panel_data <- panel_data %>%
  mutate(code = as.character(code)) %>%   # ensure same type
  left_join(
    aging_intensity %>%
      mutate(code = as.character(code)) %>%
      select(code, aging_change, high_aging),
    by = "code"
  )

panel_data <- pdata.frame(
  panel_data,
  index = c("code", "year")
)
```

Robustness model with interaction (core result)

```r
model_oops_robust <- plm(
  oops_che ~ aging_65 * high_aging,
  data = panel_data,
  model = "within",
  index = c("code", "year")
)

summary(model_oops_robust)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = oops_che ~ aging_65 * high_aging, data = panel_data,
##     model = "within", index = c("code", "year"))
##
## Unbalanced Panel: n = 184, T = 22-24, N = 4414
##
## Residuals:
##      Min.   1st Qu.    Median   3rd Qu.      Max.
## -26.71479  -2.86680  -0.14348   2.85857  28.28483
##
## Coefficients:
##                     Estimate Std. Error t-value  Pr(>|t|)
## aging_65             -2.0845     0.3574 -5.8325  5.87e-09 ***
## aging_65:high_aging   1.2304     0.3646  3.3746 0.0007458 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    164280
## Residual Sum of Squares: 157780
```

```
## R-Squared:      0.039594
## Adj. R-Squared: -0.0024299
## F-statistic: 87.1515 on 2 and 4228 DF, p-value: < 2.22e-16
```

```r
model_gghed_robust <- plm(
  gghed_che ~ aging_65 * high_aging,
  data = panel_data,
  model = "within",
  index = c("code", "year")
)

summary(model_gghed_robust)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = gghed_che ~ aging_65 * high_aging, data = panel_data,
##     model = "within", index = c("code", "year"))
##
## Unbalanced Panel: n = 184, T = 22-24, N = 4414
##
## Residuals:
##      Min.   1st Qu.    Median   3rd Qu.      Max.
## -41.93877  -3.52228  -0.16187   3.45010  28.39705
##
## Coefficients:
##                      Estimate Std. Error t-value  Pr(>|t|)
## aging_65              2.66736    0.38513  6.9258 4.988e-12 ***
## aging_65:high_aging -1.83550    0.39289 -4.6718 3.079e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    190260
## Residual Sum of Squares: 183220
## R-Squared:      0.037026
## Adj. R-Squared: -0.0051103
## F-statistic: 81.2816 on 2 and 4228 DF, p-value: < 2.22e-16
```

```r
model_shi_robust <- plm(
  shi_che ~ aging_65 * high_aging,
  data = panel_data,
  model = "within",
  index = c("code", "year")
)

summary(model_shi_robust)
```

```
## Oneway (individual) effect Within Model
##
## Call:
## plm(formula = shi_che ~ aging_65 * high_aging, data = panel_data,
##     model = "within", index = c("code", "year"))
##
## Unbalanced Panel: n = 183, T = 1-24, N = 4242
##
```

```
## Residuals:
##       Min.    1st Qu.     Median    3rd Qu.       Max.
## -37.118661  -0.930040   0.010232   0.743391  45.944763
##
## Coefficients:
##                     Estimate Std. Error t-value  Pr(>|t|)
## aging_65             1.31695    0.26025  5.0603 4.371e-07 ***
## aging_65:high_aging -0.80355    0.26533 -3.0285  0.002473 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Total Sum of Squares:    79311
## Residual Sum of Squares: 76953
## R-Squared:      0.029739
## Adj. R-Squared: -0.014266
## F-statistic: 62.1744 on 2 and 4057 DF, p-value: < 2.22e-16
```

```
library(stargazer)
```

```
##
## Please cite as:
```

```
##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
```

```
##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```
stargazer(
  model_oops_robust,
  model_gghed_robust,
  model_shi_robust,
  type = "text",
  title = "Interaction Effects of Population Aging on Health Financing Structure",
  dep.var.labels = c("Out-of-pocket (%)", "Government (%)", "Social Health Insurance (%)"),
  covariate.labels = c("Aging 65+", "Aging 65+ × High-aging"),
  omit.stat = c("rsq", "adj.rsq", "f")
)
```

```
##
## Interaction Effects of Population Aging on Health Financing Structure
## ====================================================================================
##                                            Dependent variable:
##                         ------------------------------------------------------------
##                         Out-of-pocket (%) Government (%) Social Health Insurance (%)
##                                (1)             (2)                   (3)
## ------------------------------------------------------------------------------------
## Aging 65+                  -2.085***        2.667***             1.317***
##                             (0.357)         (0.385)               (0.260)
##
## Aging 65+ × High-aging     1.230***        -1.836***             -0.804***
##                             (0.365)         (0.393)               (0.265)
##
## ------------------------------------------------------------------------------------
## Observations                 4,414           4,414                 4,242
## ====================================================================================
## Note:                                                    *p<0.1; **p<0.05; ***p<0.01
```

Across countries, increases in the elderly population are associated with a consistent shift away from out-of-pocket healthcare spending toward government and insurance-based financing. Importantly, this shift is significantly stronger in countries experiencing faster demographic aging, suggesting that health systems actively redistribute financial responsibility as aging pressure intensifies.

Create financing components(group oop, gghed, shi, and vhi)

```
hero_data <- panel_data %>%
  as.data.frame() %>%
  filter(!is.na(high_aging)) %>%
  select(
    year,
    high_aging,
    oops_che,
    gghed_che,
    shi_che,
    vhi_che
  )
```

Aggregate by aging intensity and year

```
hero_agg <- hero_data %>%
  group_by(high_aging, year) %>%
  summarise(
    Out_of_Pocket = mean(oops_che, na.rm = TRUE),
    Government = mean(gghed_che, na.rm = TRUE),
    Social_Insurance = mean(shi_che, na.rm = TRUE),
    Voluntary_Insurance = mean(vhi_che, na.rm = TRUE),
    .groups = "drop"
  )
```

Reshape to long format for ggplot

```
hero_long <- hero_agg %>%
  as.data.frame() %>%   # DROP pdata.frame attributes
  pivot_longer(
    cols = c(
      Out_of_Pocket,
      Government,
      Social_Insurance,
      Voluntary_Insurance
    ),
    names_to = "Financing_Type",
    values_to = "Share"
  ) %>%
  mutate(
    Aging_Group = ifelse(
      high_aging == 1,
      "High-aging countries",
      "Low-aging countries"
    ),
    Financing_Type = recode(
      Financing_Type,
      "Out_of_Pocket" = "Out-of-pocket payments",
```

```r
      "Government" = "Government financing",
      "Social_Insurance" = "Social health insurance",
      "Voluntary_Insurance" = "Voluntary health insurance"
    )
  )

hero_long <- hero_long %>%
  mutate(
    year = as.integer(year),
    high_aging = as.integer(high_aging),
    Aging_Group = as.character(Aging_Group)
  )
```

```r
unique(hero_long$Financing_Type)
```

```
## [1] "Out-of-pocket payments"     "Government financing"
## [3] "Social health insurance"    "Voluntary health insurance"
```

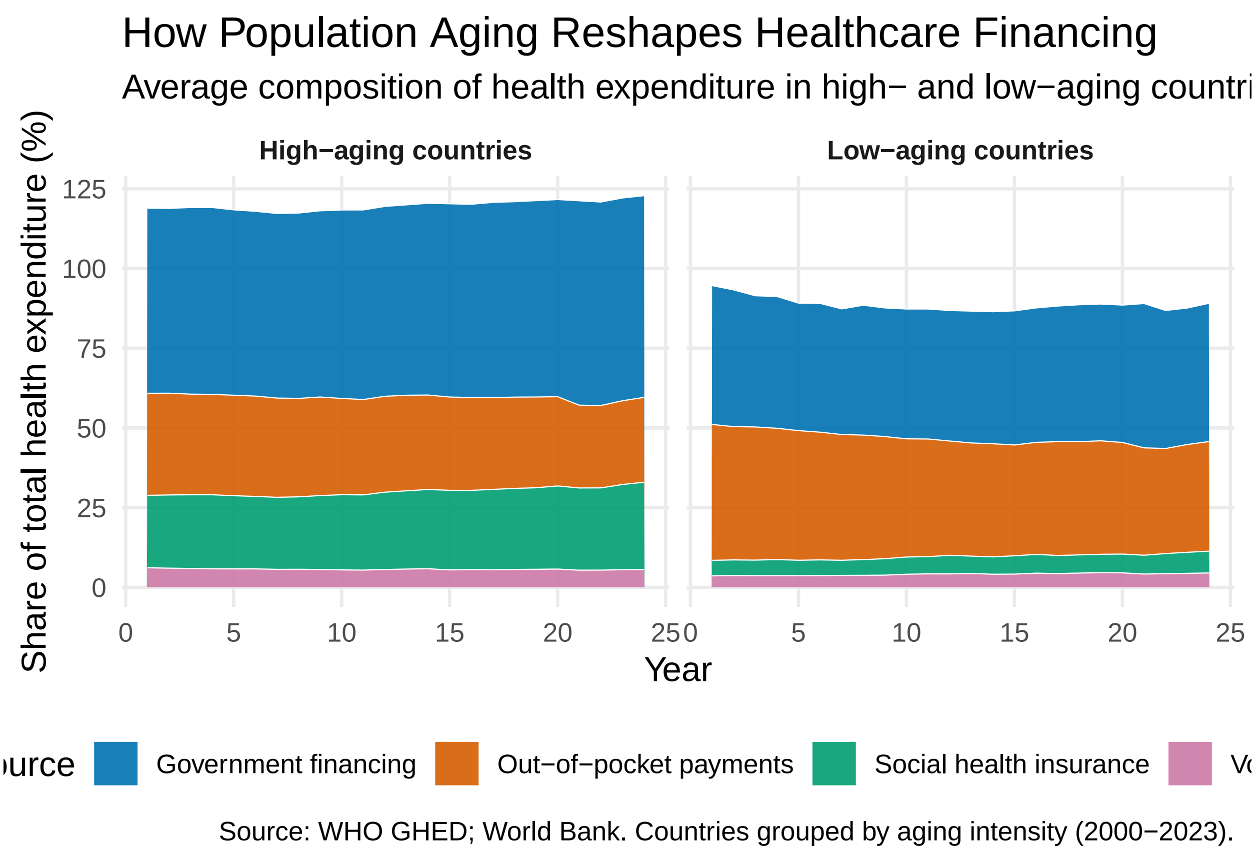
```r
levels(hero_long$Financing_Type)
```

```
## NULL
```

```r
str(hero_long$Financing_Type)
```

```
##  chr [1:192] "Out-of-pocket payments" "Government financing" ...
```

```r
ggplot(hero_long, aes(x = year, y = Share, fill = Financing_Type)) +
  geom_area(alpha = 0.9, color = "white", linewidth = 0.2) +
  facet_wrap(~ Aging_Group) +
  scale_fill_manual(
    values = c(
      "Out-of-pocket payments" = "#D55E00",
      "Government financing" = "#0072B2",
      "Social health insurance" = "#009E73",
      "Voluntary health insurance" = "#CC79A7"
    )
  ) +
  labs(
    title = "How Population Aging Reshapes Healthcare Financing",
    subtitle = "Average composition of health expenditure in high- and low-aging countries (2000-2023)"
    x = "Year",
    y = "Share of total health expenditure (%)",
    fill = "Financing source",
    caption = "Source: WHO GHED; World Bank. Countries grouped by aging intensity (2000-2023)."
  ) +
  theme_minimal(base_size = 13) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold")
  )
```


How Population Aging Reshapes Healthcare Financing
Average composition of health expenditure in high− and low−aging countri
High−aging countries
Low−aging countries
Share of total health expenditure (%)
125
100
75
50
25
0
0
5
10
15
20
25
Year
Government financing
Out−of−pocket payments
Social health insurance
Source: WHO GHED; World Bank. Countries grouped by aging intensity (2000−2023).